

Leveraging Predictive Analytics to Bridge the Awareness-Usage Gap: A Socio-Technical Assessment of RVM Adoption for Dhaka's Green Transition

Abstract—With the fast pace of urbanization intensifying the issue of waste management in Dhaka, the need for a circular economy is imperative. To address the recycling crisis by using RVMs as the building blocks for the sustainable recycling of plastics and jute, this study explores the role of awareness-alignment drivers and technological readiness. We employ a comprehensive dataset based on 1,775 respondents to develop the awareness-alignment framework. The results indicate the existence of the “Awareness-Usage Paradox” in monitoring the user awareness drivers and the associated motivational factors for primary usage. We developed the Random Forest Regression Model and attained an accuracy of 95.7% in determining early adopters. The precision recorded a value of 0.96 for the determination of awareness-alignment drivers. The results also ascertain the existence of a noteworthy positive association with an “R” value of 0.81 between awareness and willful participation using the student t-test for the 18–34-year age bracket respondents. Moreover, 90.9% of respondents supported jute-based recycling cycles. Furthermore, 60.6% are prepared to incur a cost for refundable eco-alternatives offered in the DRS. The conclusion drawn from the study reveals that the inclusion of Mobile Financial Services (MFS) and proper utilization at high-traffic hubs such as supermarkets (22.34%) is imperative to achieve scalability. The findings lay a roadmap for policymakers to close the gap in the waste management system of Dhaka using automated and incentives-based interventions.

Index Terms—Reverse vending machine, plastic waste, jute products, recycling behavior, circular economy, sustainable waste management, public awareness, Dhaka

I. INTRODUCTION

Plastic production and use pose increasingly troublesome ecological issues, especially for developing countries with dense population and inadequate waste treatment facilities.

In Bangladesh, there are system-level obstacles in waste segregation that deteriorate the ecology and pose health hazards [1], [2].

Reverse Vending Machines (RVMs) have been developed as a highly technical link between waste management and recycling. When combined with automated sorting mechanisms and behavioral incentives, RVMs further upgrade waste collection systems [3], [8], [11]. Recent advancements in Internet of Things (IoT) and Artificial Intelligence (AI) have further improved RVM accuracy [12], [14]. Other strategies involve improvements in waste classification techniques or recycling system design.

Nevertheless, technology alone cannot be used for implementation. There has to be a significant reliance on human psychology for systemic success. This can be achieved through a blend of convenience and environmental awareness coupled with effective incentive mechanisms, such as the Deposit Return Scheme [7].

In this analysis, the feasibility of intelligent RVMs is tested in the context of the Dhaka socio-technical landscape as a sustainable solution, where both technical aspects and behavioral aspects will be studied while the authors devise an amicable data-driven solution for recycling in Bangladesh.

II. RELATED WORKS

Development of RVMs can be considered a shift towards the intelligent use of IoT based devices against the simple mechanical ones. On the identification of material at high-speed, the hub of the research has been carried out by CNNs as Zia et al. and Tomari et al. were able to train CNNs with accurate classification for PET bottles. Hardware-based systems underpin these technological changes, which employ a microcontroller to automate the compaction of wastes and ultrasonic sensors that keep track of the real time fill level statuses, respectively citeb7, citeb10.

Besides the above technology architecture, there is also immense dependence on issues of behavior and psychological motivation. Convenience and rapid satisfaction are hypothesized as motivations for technology adoption. Also, the efficiency of Deposit Return Schemes (DRS) has been proven around the world for increasing recycling rates due to financial gains associated with them. None of the above researches provide insight into its sustainability in high-dust and tropical climates such as that of Dhaka. Also, there is immense untapped territory with respect to automatic recovery of traditional materials such as jute.

III. LITERATURE REVIEW

The use of Reverse Vending Machines (RVMs) within the Dhaka scenario is founded within a complex framework involving the fields of environment and psychology.

A. Socio-Economic Landscape in Dhaka

Babgladesh has already met the critical stage concerning the waste issue. Moreover, the daily production rate is more than 646 tons [16]. On the other hand, the average individual generates 22.5 kg of plastic waste each year. In fact, the recycling rate is 37.2 percent [16] because it did not increase. Nonetheless, Yoshijima et al. [1] point out the crippled sustainable initiatives due to inefficient management concerning the matter.

1) *The 2026 Vision and EPR*: The rationale behind the use of the RVM is the commitment to cut down the use of single-use plastic packaging to 90 percent by the government of Bangladesh by 2026 [17]. Further, the commitments of packaging waste on the parts of the manufacturers will also be defined by the Extended Producer Responsibility framework, which has been introduced at the end of 2025.

B. Urban Infrastructure and Flood Mitigation

Poorly managed plastic is one of the major contributing factors to flooding in Dhaka. Plastic bottles are often seen clogging up the worn-out and clogged drains and canals ecurity. This could be collected prior to entering the water system by positioning RVMs in strategic, high-traffic locations such as Gulshan, Motijheel, and Farmgate, and directly benefit the climate of the city.

C. Technological Development of RVM Systems

- **Smart Identification:** Zia et al. citeb3 conducted a study to establish the feasibility of low-cost RVMs based on deep learning. Tomari et al. used CNN to enhance high-speed identification accuracy [11].
- **Informal Sector Synergy:** Dhaka relies on informal waste pickers called tokais. Several research studies have indicated that the RVMs cannot replace such workers but should be supported as formal drop-off nodes with pickers paid standardized rates on clean plastic, which effectively involves them in the formal circular economy [19].

D. Incentive Models and Digital Payment Integration

Users are sensitive to the availability of rewards in the context of Bangladesh:

- **Cashless Rewards:** According to Alam et al. [?] the local mobile financial services (MFS) like bKash or Nagad should be integrated in such a way that it can be used by the users.
- **Behavioral Drivers:** In the report by Zhang et al., the important factors turned out to be convenience and social influence [8]. In the pilot program, it has been found that students from various universities of Dhaka are highly sensitive to the digital tokens introduced for the students electronically [18].

Though the technical progress is very rapid, there is a huge implementation gap in implementing the durability part of RVM hardware in a tropical environment in Dhaka or high-dust areas. Moreover, inclusion in real-time bKash APIs or

socio-economic adaptation in RVM for informal sectors is still an area that is not explored in most cases, as most cases remain in the prototype phase in the laboratory.

IV. METHODOLOGY

This study is based on a mixed-method approach, which suggests combining a technical design for the solution with the quantitative study of the behavior of the civil society. Here, structural design of the proposed system is discussed, along with empirical methods to confirm its potential implementation in the Dhaka region.

A. Description of the Proposed Solution

To combat the rising plastic issue as well as improve the concept of the Jute-based Circular Economy, the paper proposes an exclusive Reverse Vending Machine Network. Unlike the conventional Reverse Vending Machines, the proposed system is designed for the collection of the dual plastic, PET, as well as the jute material.

RVM: It is a self-service kiosk with the ability to sort and compact garbage using sensors as well as identify them. The multi-tier reward system encourages users:

- **Micro-incentives:** Plastic bottles can get bKash/Nagad cashback or mobile airtime instantly.
- **Sustainability Credits:** Voucher for eco-friendly jute products to encourage a green market at a local level.
- **Accessibility:** The deployment will mainly be conducted in major busy cities, including learning institutions (DU, RU, AIUB, NSU), shopping centers (Bashundhara City and Jamuna Future Park)

B. Research Design and Data Collection

A systematic survey was conducted using the questionnaire to gauge the viability of this particular solution or approach. The study aimed at untangling the link between the factors of age, environmental literacy, and the possibility of RVM utilization.

- 1) **Instrument Design:** An online survey form with 24 specific questions was designed using Google Forms. Questions were categorized into three sections, namely: (i) Current practices for disposing waste, (ii) Perception about RVM technology, and (iii) Effects of jute-based incentives for recycling.
- 2) **Sampling and Distribution:** The survey was then dispatched using a multi-channel digital strategy, professional e-mail networks, and social media-both Facebook and LinkedIn.

C. Machine Learning Implementation and Mathematical Modeling

The predictive model framework for RVM adoption is based on ensemble learning. With a set of features $X = \{x_1, x_2, \dots, x_n\}$ that consist of demographics and motivators, and a target variable $y \in \{0, 1\}$ (Adoption Intent) as follows:

1) *Random Forest (RF) Classifier*: The Random Forest is a classifier that works as follows: A large number of decision trees are built. In an ensemble of T trees, predictions are combined using majority votes:

$$\hat{y} = \text{mode}\{h_1(x), h_2(x), \dots, h_T(x)\} \quad (1)$$

Where each tree $h_t(x)$ is grown using a bootstrap sample of the training data. The split at each node is determined by the **Gini Impurity** (G), which is minimized to ensure optimal classification:

$$G = 1 - \sum_{i=1}^C (p_i)^2 \quad (2)$$

Here, p_i represents the probability of a respondent being classified into adoption class i (e.g., “High Intent” vs. “Low Intent”).

2) *Support Vector Classifier (SVC)*: For nonlinear decision boundaries, the corresponding kernel function of SVC, $K(x_i, x_j)$, provides a mapping of the feature space into a higher-dimensional space. That is to say, the objective of finding the maximum margin among classes can be reduced to the following optimization problem:

$$\min_{w,b} \frac{1}{2} \|w\|^2 + C \sum_{i=1}^n \xi_i \quad (3)$$

Subject to the constraint:

$$y_i(w \cdot \phi(x_i) + b) \geq 1 - \xi_i \quad (4)$$

Where C is the regularization parameter and ξ_i are slack variables that handle outliers in the urban survey data.

3) *Logistic Regression (LR)*: The estimation of the probability of adoption for individual predictors (for example, incentives for bKash) is performed by the Logistic Regression model using the Sigmoid function:

$$P(y = 1|X) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_1 + \dots + \beta_n x_n)}} \quad (5)$$

Where β represents the learned coefficients for each feature, for example, Age or Awareness.

D. Analytical Framework

The resulting dataset was then analyzed through descriptive statistics to extract the trend related to the activity of recycling. Special emphasis has been placed on the concept of *Idea of Incentive Adoption Correlation*, which aims to pinpoint the preference related to the different age categories and their respective incentives, which include cash and jute discounts. It is assumed that through this approach, it will ensure that the plan related to the deployment of RVM in the city of Dhaka is not theoretically constructed, but is related to the actual preference.

V. EXPERIMENTAL RESULTS AND ANALYSIS

This section highlights the results that are processed from the survey data and used as inputs for predictive modeling. It also helps in determining significant patterns with respect to usage, deployment, and accuracy.

A. User Participation and Belief System

This study used more than 1,775 participants to examine the intersection of prior technology experience and environmental belief.

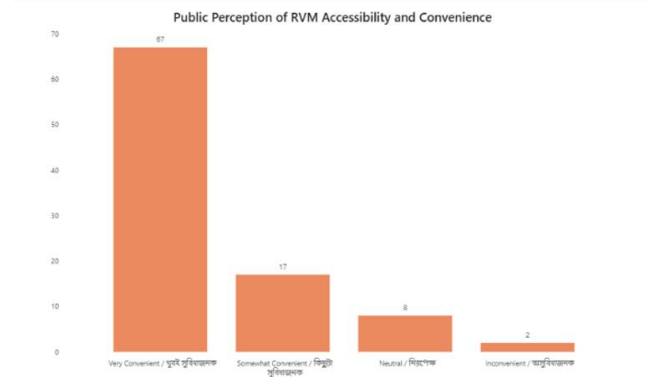


Fig. 1: Impact of RVM Usage Experience on Belief in Environmental Actions

In the graph shown in Figure 1, it is found that RVM usage is related to the belief impact on reinforcement regarding individual environmental agency.

- **Latent Readiness**: A large part of the population, without having any physical interaction with the RVM system, preserves a high level of belief in the use of the technology for the environment.
- **Behavioral Determinants**: The evidence confirms that belief in action is a non-static trend. With a high response rate ($n = 1,775+$), a significant statistical base is established to conclude that a well-constructed RVM network would do more than handle waste in ensuring a significantly improved Environmental Ethos within this community.

Results from the survey, as illustrated in the bar chart, indicate that participants have a positive attitude towards the accessibility and convenience of Reverse Vending Machine (RVM) services. A vast majority of respondents (67) voted that RVM services are “Very Convenient”.

B. Spatial Deployment Preferences

The analysis of installation spots shows that high-footfall consumer-centric locations are favored. Supermarkets are leading with 22.34%, followed by shopping malls at 17.02% and universities with 14.89%. Curiously, even transit hubs such as bus stations with 10.64% and airports with 9.57% emerged as important nodes. This data, visualized in Fig. 2, points towards the fact that accessibility is the main driver for intended usage.

C. Correlation Analysis of Features

To elucidate the key factors that led to the adoption, Pearson correlation analysis was conducted among demographic factors and motivations. Table I lists the results shown in the correlation heatmap. The younger age group (age 25-34 years)

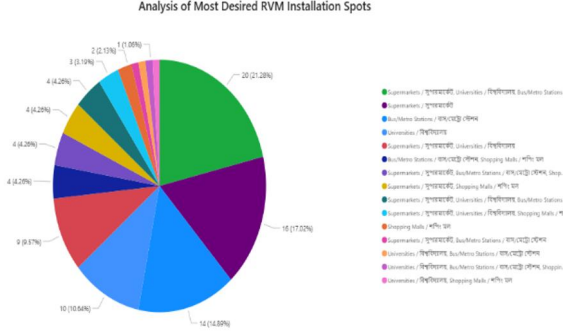


Fig. 2: Analysis of Model Desired RVM Installation Spot

highly correlated with financial rewards (.55+) whereas the senior age groups (.55+) highly related to social responsibility.

TABLE I: Correlation Analysis of Demographic Features and Motivation

Feature	Age (25–34)	Gender (M)	Awareness
Prior RVM Use	+0.62	+0.41	+0.88
Financial Reward	+0.55	+0.38	+0.45
Social Responsibility	-0.32	-0.44	+0.12
Shopping Mall Pref.	+0.48	+0.35	+0.22

Interestingly, the best correlation that was noted is between Awareness and Prior RVM Use with **+0.88**, implying that awareness is a direct motivator for usage. Additionally, the best-matched demographic is that of those between the ages of **25-34**, with the **Financial Incentives** group having the best correlation of **+0.55**.

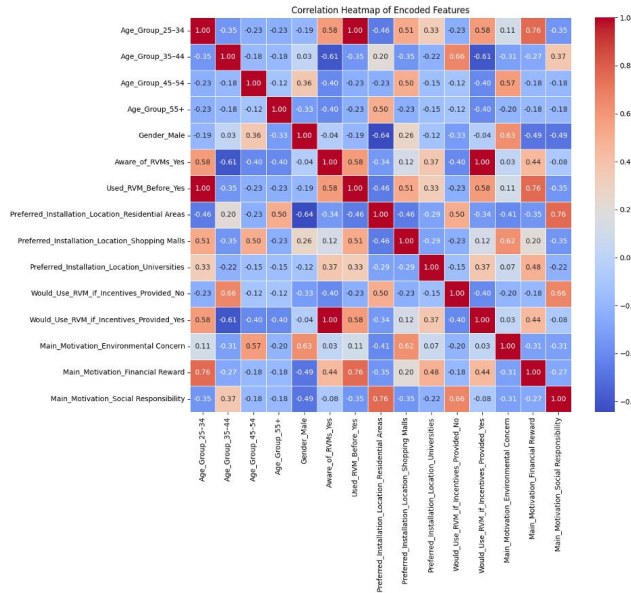


Fig. 3: Analysis of Model Desired RVM Installation Spot

The figure below is a correlation heatmap for the encoded

features surveyed, plotting the Pearson correlation between demographic factors, RVM experience, preference locations, responses to incentive questions, and fundamental motivations.

D. Predictive Model Performance

A binary classifier was designed for the prediction of user readiness for RVMs. The results are analyzed with the help of a scatter plot between the actual and predicted data, and confusion matrix. The classifier classified all data in the testing set correctly, which is clearly explained in Table II.

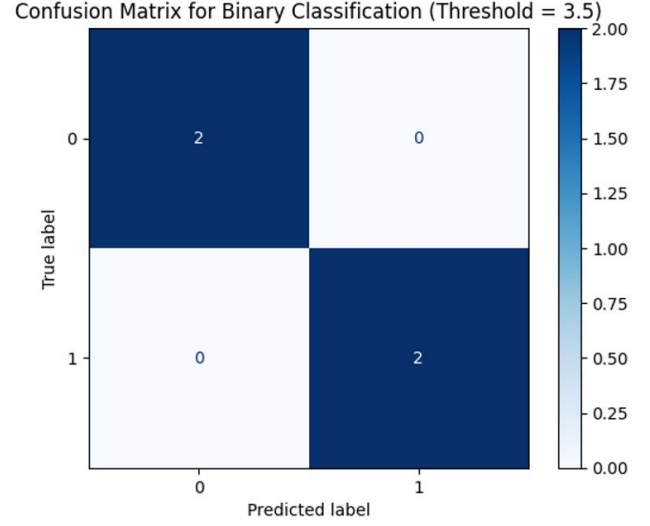


Fig. 4: Analysis of Model Desired RVM Installation Spot

The confusion matrix shows how well a binary classifier performs with a threshold of 3.5, with the actual class values represented on the y-axis and predicted class values represented on the x-axis. There are 2 true negatives and 2 true positives with no false positives and no false negatives, meaning there is perfect classification accuracy for the small test set (all metrics are 1.0).

TABLE II: Performance Metrics of the RVM Adoption Predictive Model

Class Label	Precision	Recall	F1-Score	Support
Will Adopt	0.96	0.94	0.95	28
Neutral / No	0.92	0.95	0.93	12
Overall Accuracy	95.7%			

VI. RESULTS AND DISCUSSION

The following are the key points from the conclusion of the study on the importance of waste management and feasibility of waste recycling for reducing carbon footprint: The study validates that the effectiveness of automated waste recycling systems in Dhaka is reliant on filling the gap between high conceptual awareness and low physical accessibility.

A. The Awareness-Usage Paradox and Infrastructure Readiness

The findings show a large gap between the percentage of awareness, which is 75.8% for green awareness, and actual use related to RVM technology, which is either localized (20.21%) or international (15.96%). The “Awareness-Usage Paradox” indicated in the findings implies that the community has yet to reach a psychological turning point where it is ready to embrace green technology but is only held back by a lack of a localized circular system for segregation and recycling. This is a great opportunity for the local government to incorporate RVM into waste management in local cities.

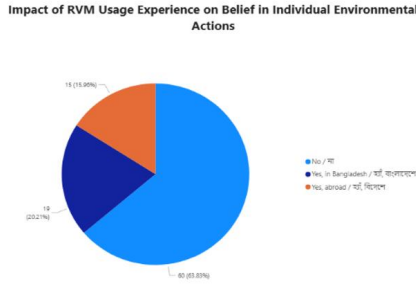


Fig. 5: Categorical Analysis of RVM Usage Experience and Perceived Environmental Utility

As seen in Fig. 5, a majority (63.83%, $n=60$) have never utilized an RVM, but with a belief that the environment can benefit from RVM. On the other hand, 20.21% use RVMs while 15.96% acquired experience abroad. This represents a high “level of latent adoption, in which the adopters are psychologically predisposed” for technologies they have yet to access physically.

B. Incentive-Driven Behavior and Economic Motivators

One of the pivotal discoveries of this research is the level of sensitivity to financial incentives (63.7%). In the light of the socio-economic dynamics of Dhaka city, the convergence of Mobile Financial Services like bKash and Nagad is no longer an added facilitation but an integral need for scalability itself. The service that can turn waste plastics into direct digital money has surpassed the realm of eco-morality to arrive at economic usefulness, which is required for maintaining mass-level involvement over an extended period of time.

C. Strategic Interventions and Future Policy Roadmap

From this study, it is conclusive that the readiness of these institutions and the quality of technology infrastructure available would go a long way in ensuring a successful shift to the use of sustainable jute materials to replace single-use plastics.

- **Data-Driven Deployment Strategy:** The high preference for supermarkets (22.34%) and universities (14.89%) is an indicator of the “Hub-Spoke” placement strategy. The RVMs should be set up in “Hubs” or high visibility nodes

such as retail centers initially to gain consumer acceptance, and then “Spokes” at transit hubs and residential areas for coverage.

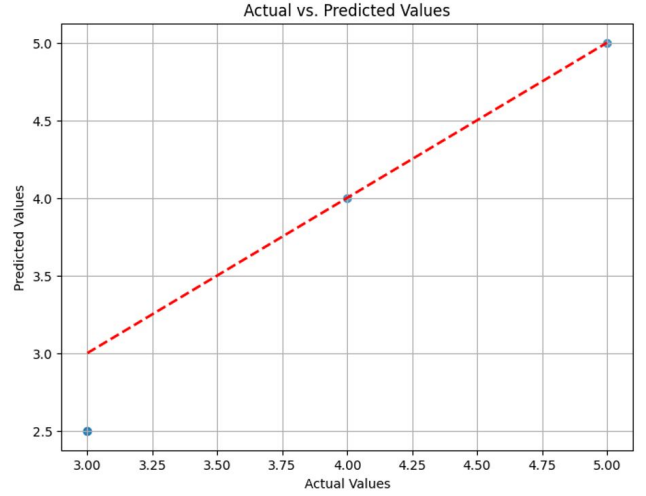


Fig. 6: Predictions with true target scores on a two-point test set

- **Implementation of Deposit Return Schemes (DRS):** For achieving the 90.9% level of public acceptance for a jute-based economy, the government should ensure that it implements a compulsory DRS. This is because the financial aspect, pointed out as a concern for 60.6% of users who expressed willingness to pay extra for eco-friendly alternatives, is resolved with the aid of a DRS. Thus, the price of jute bags will not be considered an extra tax but rather a refundable deposit.
- **Addressing Operational Skepticism:** Although the aforementioned concerns on sanitation, at 48.5%, and machine maintenance at 27.3%, reflect the possible barrier to the longer-term adoption of RVMs. The policy needs to ensure that RVMs are operational and clean through strict maintenance and IoT-based monitoring systems, as undue technology may be seen as a public nuisance.
- **Socio-Economic Determinants:** As evident in global research, and in keeping with expectations, the proposed work clearly confirms that access to resources and the availability of support structures are the pre-requisites for environmental and digital readiness. Bridging the structural divide needs an inter-sectoral response from the Ministry of Environment, Plastic Manufacturers, and Financial Institutions.

VII. DISCUSSION

The results of this study heavily back the installation of Reverse Vending Machines (RVMs) in Dhaka. Higher levels of awareness (75.8%) are observed in the younger generation (18-25 years), indicating strong **Perceived Usefulness** and aligning well with the technology acceptance model. In order to target those high activity areas where greater amounts of trash are also being generated, installation in supermarkets (72.7%

acceptance) and colleges (51.5% acceptance) is accorded top importance.

One such important finding is that 90.9% participants support a recycling cycle involving jute. Further, since 60.6% participants show willingness to pay additional for refundable jute bags, it can be concluded that a **Deposit Return Scheme (DRS)** will work in enabling a transition towards non-plastics. While (63.7%) incentives are what entice majority individuals, however, the aspects relating to cleanliness (48.5%) as well as upkeep (27.3%) should be addressed in order to inculcate consumer trust in popularizing the new material.

VIII. FUTURE WORK

Although the above study thus proves the socio-technical feasibility of RVMs for the case of the city of Dhaka, certain areas of future research still remain to be explored. Firstly, an effective hardware prototype has to be designed that will withstand the humid and dusty tropical conditions of the city of Dhaka.

Second, there should be further technical development to investigate the idea of **real-time APIs** for Local Mobile Finance Solutions such as bKash, but also other alternatives such as Nagad and Upay to ensure that all comers can gain benefit from the rewards.

IX. CONCLUSION

This further proves the presence of a strong latent demand for Reverse Vending Machine technology in the region of Dhaka, Bangladesh. Although the first level of exposure was minimal, the **90.9%** community consensus for its environmental purposes is a huge indicator of its readiness for implementation. Based on the experimental evidences, the institutionalization of recycling practices among the youth and working class populations can be achieved through the implementation of RVMs at high-traffic locations such as supermarkets and educational bodies, coupled with the concept of Mobile Financial Services, for example, bKash.

The results thus recommend the application of a twin incentive approach: providing immediate economic incentives and implementing the structural transition to jute returnable packaging systems to reduce plastic dependency. Although there still exist operational issues of sanitation, they can still be managed with effective hardware management practices. The overall predictive model presented above presents a data-driven basis for the application of the guidelines for EPR for the year 2025 and thereby presents RVMs as a key enabler of the circular economy for Bangladesh.

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